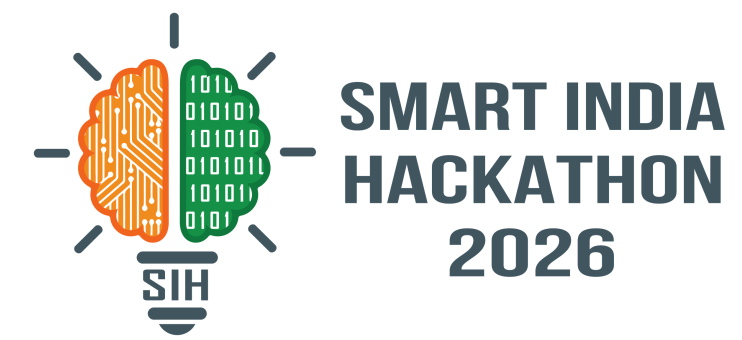


SMART INDIA HACKATHON 2026



SURANG-SUTRA

The rescue robot-drone that finds, triages & talks to people trapped where GPS & signal die

सुरंग (tunnel) + सूत्र (thread) · जहाँ सिग्नल खत्म, वहाँ सूत्र शुरू (where signal ends, SUTRA begins)



ROBOT + DRONE

rolls through tunnels, flies over debris



LAYS ITS OWN LIFELINE

₹900 relays dropped at every bend



FINDS SURVIVORS

heat · sound · breath · ID tag



REMOTE TRIAGE

breath · pulse · air → a doctor talks

≈ ₹3 lakh field unit · 16–24× cheaper than imported caged drones

Problem Statement ID - 26218

Problem Statement Title-

Student Innovation-There is a need to design drones and robots that can solve some of the pressing challenges of India such as handling medical emergencies, search and rescue operations, etc.

Theme- Robotics and Drones

PS Category- Hardware

Team ID- 121295

Team Name- Team Palanteen

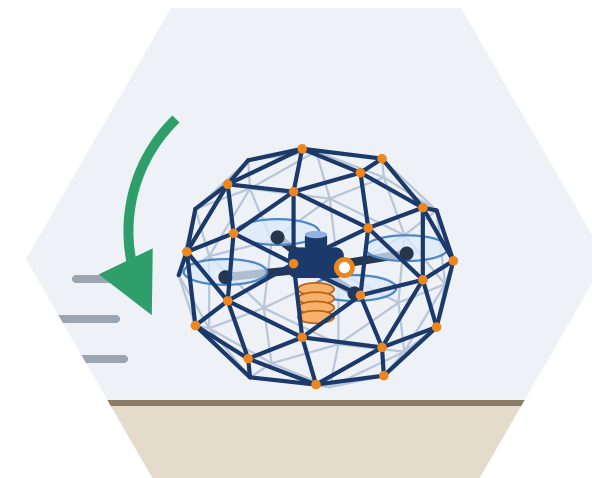
Thakur Shyamnarayan Engineering College



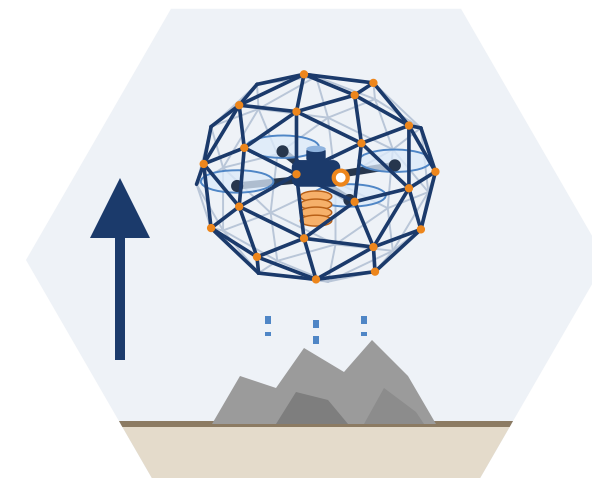
 **LIVE 3D MISSION SIMULATION**

surang-sutra-simulation.vercel.app

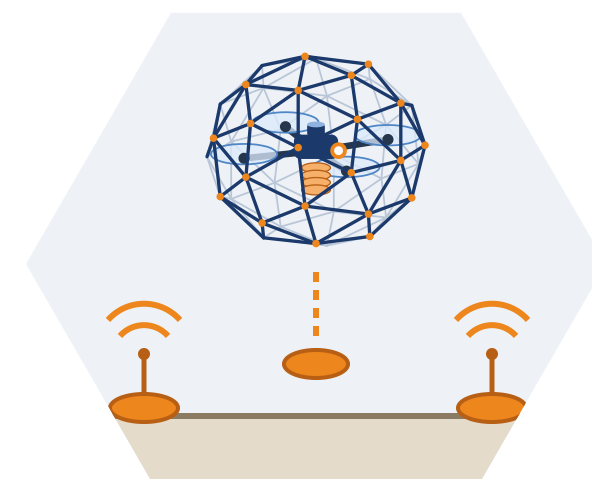
Working software prototype (simulation, not hardware)



1 ROLLS through the tunnel



2 FLIES over debris & water



3 DROPS a SUTRA relay



SURANG-SUTRA

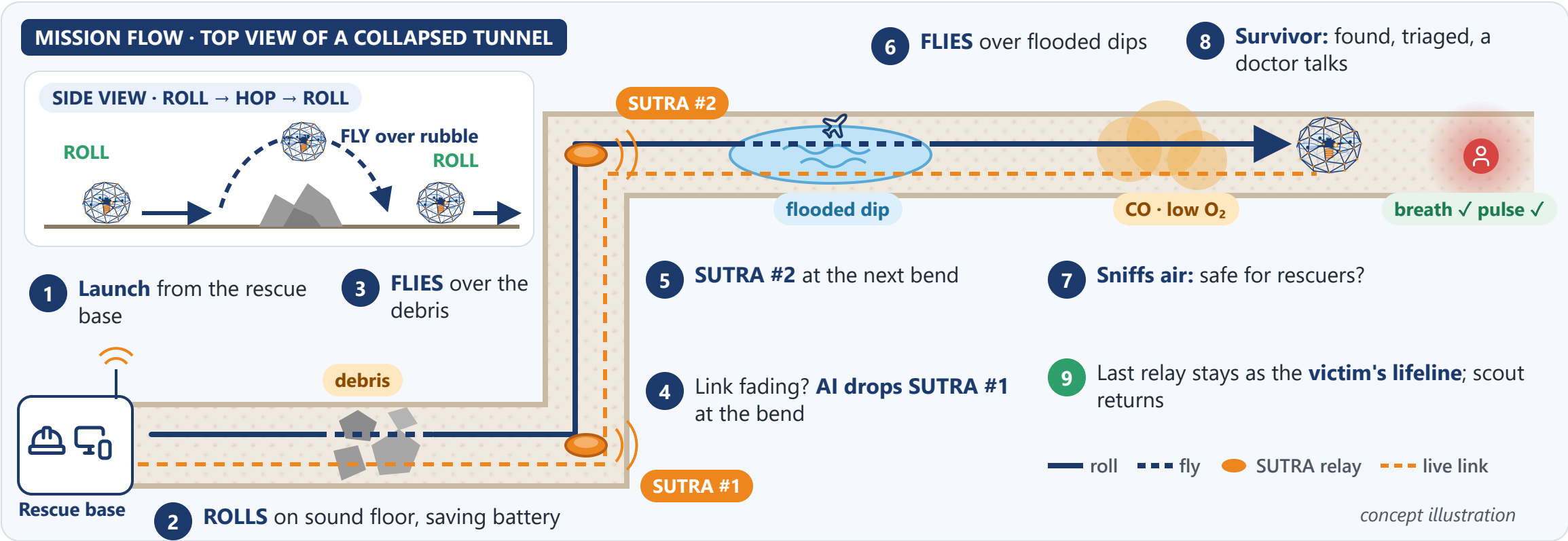
Proposed Solution (Describe your Idea/Solution/Prototype)

SURANG rolls in, flies over debris & water, **finds & triages** trapped people and drops **₹900 SUTRA relays**, so video, vitals & voice reach rescuers and doctors first.

 **SEARCH & RESCUE** tunnel, mine & building collapses, floods

 **MEDICAL EMERGENCY** remote triage + a doctor's voice before extrication · the 9-step mission below

Detailed explanation of the proposed solution



TECHNICAL APPROACH

Technologies to be used (e.g. programming languages, frameworks, hardware)

INSIDE SURANG · SIDE CUTAWAY

1 Roll axle crosses the direction of travel: the cage turns on a roller race while the sensor core stays level.

- 1 **GEODESIC ROLL-CAGE**
GFRP ribs, TPU joints, one axle
- 2 **4 × 7-INCH ROTORS**
same thrust flies and rolls it
- 3 **SENSOR SPINE**
LiDAR · thermal · IR · radar · gas
- 4 **DUAL BRAIN**
Pixhawk safety + Pi 5 & Hailo AI
- 5 **6S BATTERY**
≈ 2 kg all-up, T/W ≥ 3 (target)
- 6 **SUTRA CASSETTE**
5 relays · 1 servo · 1 per drop

HARDWARE

- **Pixhawk 6C Mini** flight controller
- **Raspberry Pi 5 + AI HAT+ 2** (Hailo-10H, 40 TOPS)
- **Unitree L2** 3D LiDAR
- **FLIR Lepton 3.5** thermal + **IMX327** IR night vision
- **60 GHz radar** (breathing) + **gas**: CO, O₂, H₂S, CH₄, CO₂

FIRMWARE & RADIO

- **ArduPilot** (C++): roll/fly modes, failsafes
- **ESP-IDF** (C) on every relay
- **Wi-Fi mesh** fast lane + **ESP-NOW Long-Range** slow lane
- **Wi-Fi FTM** ranging → Breadcrumb GPS

SUTRA relay: ESP32-C3, ≈ ₹900; the last = victim's lifeline

AI & SOFTWARE

- **ROS 2** (Python / C++)
- **FAST-LIO2** 3D LiDAR SLAM
- **YOLO11** on thermal + IR, pre-trained on **synthetic data from our own 3D sim**
- **Triage AI**: camera pulse (rPPG) + radar/thermal breathing → **START colour**
- **Air check**: safe-to-enter CO · O₂ · CH₄

COMMAND CENTRE

- **Three.js** 3D mission workspace **BUILT ✓**
- **Mission replay** + product explorer **BUILT ✓**
- **MQTT** telemetry + black-box logs
- **Bhashini** speech AI, 22 Indian languages **add-on**

Methodology and process for implementation (Flow Charts/Images/ working prototype)

HOW IT WORKS · SENSE → THINK → ACT → CONNECT → COMMAND

SENSE

- 3D LiDAR
- Thermal + IR night vision
- Breathing radar
- CO, O₂, CH₄, H₂S

THINK

edge AI on SURANG

- Live 3D map
- Survivor fusion
- Triage AI
- Air check

ACT

- Roll ⇌ fly switch
- Drop relay at a bend
- Perch, props idle

CONNECT

- Relay chain, hop by hop
- Alerts first, video last
- Voice on the slow lane

COMMAND

- Map + hazard zones
- Triage colour + doctor talk
- **Commander decides**

AI RELAY-DROP BRAIN · ONE-SERVO RELEASE

① Link fades at a bend → ② HOLD at the apex → ③ ARM next relay → ④ RELEASE one → ⑤ VERIFY new hop ✓

⊗ No new hop? Fall back to the Long-Range lane and step back to the last good link.

1 REST

lower pin holds the stack

2 CAPTURE

upper pin grips the rest

3 RELEASE

only the bottom relay drops

4 INDEX

next relay moves down

One servo, one cam, both pins: exactly one relay per drop; it lands upright and pops up its antenna.

WORKING PROTOTYPE · SOFTWARE · LIVE NOW

3D tunnel · mission replay · product explorer. **116 automated checks passing.**

FEASIBILITY AND VIABILITY

Analysis of the feasibility of the idea

FEASIBILITY AT A GLANCE

TECHNICAL

✓ OFF-THE-SHELF

- Every part is **off-the-shelf**
- Open stack: ArduPilot · ROS 2 · FAST-LIO2
- Software prototype live**

₹ ECONOMIC ✓ LOW COST

- ₹1.5 L SIH prototype (team BOM)
- ≈ ₹3 L field unit (est.)
- Sensors return; only ₹900 relays stay

OPERATIONAL ✓ FITS

- 1 operator + 1 laptop**
- Launches from the rescue base or van
- Search & rescue** + triage modes

SAFETY PATH PLANNED

- Gas-free trials** first
- Doctor-in-the-loop** triage
- Drone Rules 2021 · Ex-proof → **DGMS**

COST VS IMPORTS

16–24× cheaper: our field unit vs the leading imported caged drone

Elios 3 Enterprise	₹71.5 L
Elios 3 Light	₹49.5 L
SURANG field unit	≈ ₹3 L (est.)
SURANG SIH prototype	₹1.5 L

Five SUTRA relays = ₹4,500, robot-dropped, vs a ₹5.5 L hand-placed Elios extender (~120×)

Flyability price list v3.6 (2023) at ₹95.96/US\$ · prototype = team BOM ₹1,29,515 + 15 %

Potential challenges and risks

Strategies for overcoming these challenges

 **Radio dies around bends**
≈ 24 dB lost at a single bend

→ **AI drops a SUTRA relay at the bend apex** · Long-Range fallback · black-box keeps full data

designed-in

 **Contactless vitals can mislead**
dust, angle, movement

→ **Fuse radar + thermal + camera**, read while perched, props idle · AI only suggests; a doctor confirms

designed-in

 **Short drone endurance**
caged drones fly about 10 min

→ **Roll-first mission**: rolling cages showed ~4× range in research; fly only over obstacles

designed-in

 **Voids tighter than the cage**
500 mm cage vs dense rubble

→ **Tunnels, galleries, stairwells & voids ≥ 0.6 m** · SURANG-mini on the roadmap

roadmap

 **Relay jams or double-drops**
two pucks fall together

→ **Two-pin escapement + break-beam exit check** · one servo, no motor in the relays

designed-in

 **False survivor alarms**
hot rock, noise, echoes

→ **Multi-sense fusion + confidence score** · a human confirms with one tap

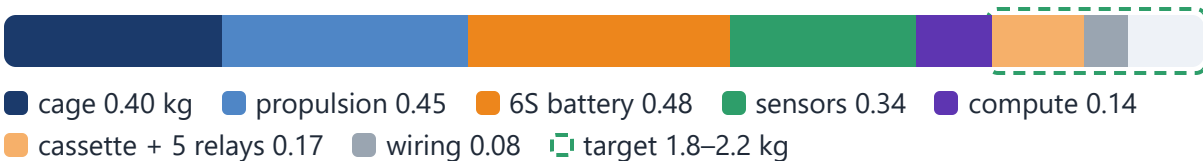
designed-in

 **Ignition risk in gassy mines**
motors and battery near CH₄

→ **CH₄ interlock** (0.75 % turn back, 1.25 % land) · gas-free trials · Ex-proof → CIMFR → DGMS

roadmap

MASS & POWER BUDGET · PLANNING ESTIMATE



≈ **2.1 kg all-up** · thrust ≈ 4 × 1.7 kg → **T/W ≈ 3.3** · flight 8–10 min, **roll-first mission 25+ min** (targets; weighed at the Oct bench test)

ROADMAP · BUILD TIERS

- NOW · SEP 2026 ✓** Live 3D simulation + mission replay; design frozen
- OCT–NOV 2026** Bench proofs: 100-cycle relay release, bend radio test, thrust & roll test
- DEC 2026 · SIH FINALE** Core tier: roll-fly cage, 2 live relays, thermal + triage demo, 3D map, command centre
- 2027** Add-ons: radar vitals, geophone, voice, AI brief · mock-collapse trials with an SDRF & a medical college (proposed)
- 2027+ · PHASE 2** e-nose, SURANG-mini for tight voids, digital twin · pilots with NDRF, tunnel builders & mine rescue (proposed)

IMPACT AND BENEFITS

Potential impact on the target audience

TODAY: RESCUERS GO IN BLIND

- ✗ Silkyara 2023: first sight of 41 trapped workers on **day 10**, via a medical endoscope
- ✗ Radio & GPS **die under rubble & round bends**
- ✗ The first "sensor" in bad air is **a human**
- ✗ Victims wait **untreated**: no doctor can see or hear them

VS

WITH SURANG-SUTRA: SEE FIRST

- ✓ **First eyes inside in minutes** (target); no person goes first
- ✓ **Remote triage**: breath, pulse & air sent to a doctor
- ✓ **Relay trail** keeps video & voice live
- ✓ Survivors **found, triaged, spoken to** in their language

concept visualisation

WHO GAINS



NDRF / SDRF
collapses & floods



Fire & emergency services
first on scene in cities



Doctors & hospitals
triage before extrication



Tunnel & metro builders
NHIDCL · rail · metro



Mine-rescue stations
Coal India · DGMS



Trapped people & families
faster, informed rescue

WHY INDIA, WHY NOW

8,756

deaths in structure collapses, 2018–22: **~5 a day**

Day 10

first sight of Silkyara's 41 trapped workers (2023)

498

sewer & septic deaths, 2019–Jun 2026: **humans go in first**

Sources: NCRB ADSI (Factly) · ANI, Nov 2023 · MoSJE Lok Sabha reply, Aug 2026

Benefits of the solution (social, economic, environmental, etc.)

SOCIAL

0 rescuers
in the first, deadliest entry

- Humans follow a **mapped, air-checked route**
- A doctor's voice** reaches the trapped
- Triage starts** before extrication

ECONOMIC

≈ ₹3 L
field kit vs ₹49.5–71.5 L import

- One kit **per NDRF / SDRF team**
- Sensors **come home**; only ₹900 relays stay
- One tool for **collapse, tunnel & mine** calls

ENVIRONMENTAL

Reuse
not throw-away sensor nodes

- Rechargeable, recoverable** relays
- Victim located: **less blind drilling** in debris
- Repairable** cage sectors

STRATEGIC

Make in India
open, indigenous design

- 22-language voice** via Bhashini
- One scout for **collapses, tunnels & mines**
- Scales to **every NDRF battalion & SDRF**

SURFACE COMMAND CENTRE **BUILT TODAY · SIMULATION**

3D mission Mission replay 3D Product Explorer Reference mine scan

MISSION SURANG / FORWARD CAMERA SURANG-01 LIDAR / DISTRICT PLOT SWEEPING · 13 Hz SURANG-01: NOMINAL

MISSION STATUS Elapsed 00:00 Link 96% Relays 0/5 Surveyed 0% Vehicle Nominal

MISSION CONTROLS MAN, AUT, FLY, LEAD, SWP, LID, THB, MOD, Start recording, CAM

MISSION DISPLAY ROLL PORTAL DRIFT TAKE CONTROL

MISSION METRICS RETRY, SWEEP, TRACK, RELAY, BATTERY, 100%, 96%, 0.0, 0.0, 0.00 m/s, 0.00 m, 2.45 m, 0.00 m/s, 0.00 m/s, 0.00 m/s

MISSION ACTIONS AIR AT VICTIM CAUTION · CO 60 ppm SURVIVOR #1 · YELLOW breath 24/min · pulse 110 Confirm Talk AI BRIEF · 60 s Survivor 42 m past bend 2, conscious. Send O₂ first.

RESEARCH AND REFERENCES

Details / Links of the reference and research work

RESEARCH → OUR DESIGN

every innovation is anchored in published work

HyTAQ · ICRA 2013 [1]
A rolling cage gives 4× range, ~6× operating time vs flying
→ SURANG rolls by default, flies only over obstacles
ROLL-FLY

Appl. Sci. 2026 · mine tunnels [2]
≈ 24 dB lost at a single bend in a mine tunnel
→ a SUTRA relay is dropped at each bend apex
RELAY AT EVERY BEND

CERBERUS · DARPA SubT [3]
Winning robots dropped Wi-Fi breadcrumbs underground
→ our drop brain adds LiDAR bend prediction
AI RELAY-DROP BRAIN

Sci. Rep. 2025 · coal-mine SLAM [4]
Dropped tagged targets cancel map drift in long roadways
→ every relay carries an AprilTag + radio ID
BREADCRUMB GPS

Espressif ESP32-C3
Built-in Wi-Fi ranging (1–2 m) + Long-Range mode
→ relays range each other with no extra parts
₹0 UPGRADE PER RELAY

Fraunhofer 2023 · NIOSH [5]
Drones hear cries for help; geophones hear tapping
→ rotors idle while perched for clean audio
PERCH & LISTEN

Anal. Chem. 2018 · YOLO 2025 [6][7]
Thermal AI finds people; sensors smell entrapped victims
→ fused cues, confirmed by a human
MULTI-SENSE SEARCH

Bioengineering 2023 · drone rPPG [8]
Pulse from drone video at 5 m: good enough for triage
→ contactless triage colour; a doctor confirms
REMOTE TRIAGE

Codec2 · Bhashini (MeitY) [9]
Voice in 3.2 kbps; speech AI in 22 Indian languages
→ doctor ↔ victim voice rides the relays
TALK-THROUGH

Flyability price list 2023 [11]
Elios 3: US\$51.6k–74.5k + \$5.7k range extender
→ our field unit targets 1/16–1/24 of that
≈ ₹3 L FIELD UNIT

REFERENCES · CLICK ANY LINE

[1] [HyTAQ hybrid rolling–flying robot · ICRA 2013](#)

[4] [Multimodal SLAM in coal mines · Sci. Rep. 2025](#)

[7] [YOLO miner detection, thermal · MME 2025](#)

[10] [Silkyara: endoscope makes first contact · ANI 2023](#)

[13] [498 sewer & septic deaths since 2019 · BS 2026](#)

[2] [Bend loss in mine tunnels \(LoRa\) · Appl. Sci. 2026](#)

[5] [Fraunhofer "drone with ears" · 2023](#)

[8] [Heart rate from drone footage in disasters · 2023](#)

[11] [Flyability Elios 3 price list · 2023](#)

[14] [Coal Mines Regulations 2017 · Reg. 153 \(DGMS\)](#)

[3] [CERBERUS, DARPA SubT winner · 2022](#)

[6] [Sniffing entrapped humans · Anal. Chem. 2018](#)

[9] [Bhashini Indian-language speech AI · MeitY](#)

[12] [Structure-collapse deaths, NCRB ADSI · Factly](#)

[15] [SIH 2026 problem statements · PS 26218](#)

Live software prototype (simulation): [Mission replay](#) · [3D tunnel scene](#) · [Product explorer](#)

Concept renders and simulations are labelled as such; hardware performance figures are design targets to be measured in the Oct–Nov 2026 bench tests.